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COI Name (Institution):	Chapes, Stephen Ph.D. (Kansas State University)		
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The spaceflight environment influences immune cell distribution, ex vivo responses to antigen, and cytokine expression. However, few studies have characterized the impact of the spaceflight environment on the host immune response following an in vivo immune challenge, particularly those involving B cells and antibody production. The International Space Station (ISS) provides an ideal laboratory within which investigators can explore this unique environment and its impact on the host response. Through the use of a mouse model (with an immune system that closely parallels that of humans), we will establish a link between exposure to the spaceflight environment and decrements in immune system activation. This study is significant because it will directly test the hypothesis that host immunological health will be compromised in space.

 On-orbit Flight Activities, Procedures, and Timeline

Mice will receive health checks upon unloading and transfer from the Transporter to the Habitat after capsule docking no later than 24 hrs after hatch opening. Animal health will then be monitored daily via continuous video recording from one camera at a time from all of the Habitats on-orbit (including other investigations which may be running concurrently) and down-linked to Ames science staff. Ground personnel shall have the option of programming or manually controlling which camera is being down-linked and recorded at any given time. Water levels will be checked every 3-4 days. After a minimum of 14 days after animal transfer from the Transporter to the Habitat, all 20 mice will be moved to the MSG and receive intraperitoneal (i.p.) injection of vehicle (5 mice), Tetanus Toxoid (TT) vaccine (5 mice), CpG ODN adjuvant (5 mice), or TT vaccine + CpG ODN adjuvant (5 mice). The dose of TT and/or CpG ODN will be 10 microg/ mouse and 40 microg/ mouse, respectively. Pre-mixed doses will be prepared preflight in 100 microl saline and stored at -20 to -30°C until the appropriate time point. All injections and blood draws will use a 1 mL 26G 3/8” tuberculin syringe. All injections will be i.p.

Before use, the CpG adjuvant must be stored at -20°C or colder until use and the TT vaccine must be stored at 4°C. At 14 +/- 2 days after injection, mice will be transferred to the MSG and sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Following spleen collection, the remains of the mouse carcass will be wrapped in tin foil, put in a zip-lock bags and placed in a -80°C environment (or colder) to preserve the tissues for future biospecimen sharing. Blood must not be allowed to sit at ambient temperature for longer than 1 hour post-collection; if operational requirements require the blood collected to be stored in excess of 1 hour before centrifugation, the blood must be transferred to a 4°C environment until centrifugation, for a period not to exceed 4 hours. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to Earth. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable.
 <table border=1 cellspacing=0 cellpadding=0 align=center width=672> <tr > <td width=91> Time </td> <td width=252> Activity & Estimated Time Required </td> <td width=76 > Thermal Req. </td> <td width=253 > Comments, Location & Special Considerations </td> </tr> <tr > <td width=91 > Launch to L + 45 days </td> <td width=252 > Animal health checks via video monitoring; water level checks every 3-4 days </td> <td width=76 > 20 – 30°C </td> <td width=253 > Video downlink required for health checks </td> </tr> <tr> <td width=91 > Transfer to Habitat + 14 d </td> <td width=252 > All 20 mice will be transferred to habitat </td> <td width=76 > 20 – 30°C </td> <td width=253 > Requires transfer to and from Habitat with MTB; </td> </tr> <tr > <td width=91 > Injection + 14d +/- 2 d </td> <td width=252 > At specific time points, all 20 mice will be removed from AEM and given appropriate injection </td> <td width=76 > 20 – 30°C / -80°C </td> <td width=253 > Required transfer to MSG for injections </td> </tr> <tr > <td width=91 > Dissection + 0 d </td> <td width=252 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen. Blood and spleen samples kept at -80^oC until delivered to laboratory of PI </td> <td width=76 > -80°C </td> <td width=253 > Required transfer to MSG for dissection, processing of blood in the centrifuge, and storage of samples at -80°C for up to 6 months. </td> </tr> </table>

  Hardware Housekeeping Operations: Crew-performed

Daily hardware checks will occur. Habitats will be kept at a regular 12hr:12hr day:night light cycle to match the day/night programmed to be synchronous with the crew’s day/night cycle upon arrival. Water refill of habitats shall be performed. To prevent rodent food bars from becoming too hard or soiled, they will be replaced with a fresh food bar plate (from 4°C storage) at a time to not exceed 30 days after the introduction of mice to a Habitat.

Task Description:

Ground Activities Accompanying On-orbit Activities and Timeline

Ground controls will be run following spaceflight environmental parameters in orbital environmental simulator(s) (OES) chambers at KSC ACF. Delay between launch and start of environmental controls may be up to 8 days, depending on the logistics and capability of the telemetry system. Mice will be housed in flight Habitats with RFB diet and lixit water to match on-orbit groups. The operations done on ground will be conducted to match the same relative time points during the day/night cycle of the animals on-orbit, so that procedures on ground control mice are conducted at the same points in the animals’ circadian rhythm.

Each experimental group for ground controls will be doubled to improve the statistical significance of the experiment with a total of 40 mice (4 groups of 10). Therefore, 2 Transporters will be required for the flight in orbital environmental simulator(s) (OES) that have been programmed with telemetry data from the flight. At the same relative time point (accounting for delay in starting ground controls after flight) the mice are transferred from the Transporter to the Habitat, the 40 ground control mice will be transferred from the 2 Transporters into 4 Habitats, and maintained in OES programmed with telemetry data from the ISS. OES shall be programmed to mimic the following conditions experienced by the Transport and then the Habitat: CO2, O2, temperature, and humidity.

Injection and dissection must be performed as close as possible to how they are done on-orbit. In particular, the time of day that mice are sacrificed and dissected on the ground should match the same point in the circadian rhythms of the mice on-orbit. However, dissections or ground control mice can be conducted on standard laboratory benches/tables

with chairs that are more ergonomic to the operators. For ground controls, dissection pins or needles may be used to pin down mice for dissection, as can spray bottles of wetting the mice before dissection. Flight hardware is required when it may affect the tissue dissection or preservation, such as having flight cold blocks in mini cold bags for tissue freezing.

Mice will receive health checks daily via video in the Habitat and water levels will be checked every 3-4 days. After a minimum of 14 days in the OES, all 40 mice will receive an i.p. injection of vehicle (10 mice), TT vaccine (10 mice), CpG adjuvant (10 mice), or TT vaccine + CpG adjuvant (10 mice). At 14 +/- 2 days after injection, mice will be sacrificed via an i.p. injection of 100/10 mg/kg ketamine/xylazine mixture, followed by exsanguination by cardiac puncture. Immediately following the blood collection, the spleens from the mice will be removed, sectioned into halves, and placed in a frozen environment (minimum temperature -80°C) until returned to Earth. Duration for which blood is stored at ambient or 4°C shall be matched as closely as possible to the timelines the same procedures will be performed on-orbit. Blood collected will be separated into solid and serum/plasma fractions by centrifugation and frozen at -80°C until returned to the laboratory of the PI. Storage times for both spleen and blood of up to 6 months at -80°C are acceptable.

<table class=MsoNormalTable border=1 cellspacing=0 cellpadding=0 align=center width=672 > <tr > <td width=145 > Time </td> <td width=186 > Activity </td> <td width=66 > Thermal Req. </td> <td width=275 > Comments, Justification, Special Considerations </td> </tr> <tr > <td width=145 > Beginning of ground controls to +45 days </td> <td width=186 > Animal health checks; water level checks every 3-4 days </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Video downlink required for health checks </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > At specific time points, all 40 mice will be removed from AEM and given appropriate injection </td> <td width=66 > 20 – 30^oC </td> <td width=275 > Transfer from cages to a regular lab bench and back w/ standard cages. </td> </tr> <tr > <td width=145 > TBD, depending on delay in ground controls and on-orbit timelines </td> <td width=186 > Mice euthanized, blood drawn and centrifuged and spleen sectioned and frozen </td> <td width=66 > -80^oC </td> <td width=275 > Processing of blood in a centrifuge, and storage of samples at -80^oC </td> </tr> </table>

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

According to the National Institute of Aging (NIA), “Life expectancy nearly doubled during the 20th century with a ten-fold increase in the number of Americans age 65 or older.” Furthermore, “there are approximately 35 million Americans age 65 or older, and this number is expected to double in the next 25 years.” Consequently, there is a rapidly growing need to understand the mechanisms underlying age-associated illness. Indeed, in the last decade, research has repeatedly demonstrated an impact of aging on immune function. Low-grade increases in circulating inflammatory cytokine secretion are often associated with senescence. Furthermore, studies suggest that these disruptions are likely due to decrements in cellular repair mechanisms and/or increases in antigenic load. Given this disruption in immune function, it is perhaps not surprising that many of the diseases associated with aging — such as type 2 diabetes, rheumatoid arthritis, Parkinson’s disease, Alzheimer’s disease, cardiovascular disease — have strong inflammatory underpinnings.
There are strong indications that the physiological effects of spaceflight are also similar to those seen in senescence. For instance, evidence suggests that age-associated enhancements in inflammatory cytokines are linked to changes in adipose tissue and energy balance. Similar links between metabolism, adipose tissue, and inflammation have also been shown to manifest in astronauts and animal models after exposure to spaceflight. We propose to use long-term exposure to the spaceflight environment to model the effects of aging on immune function. Although these initial studies involve mice, the next obvious step in this work will be to work directly with astronauts. Because astronauts tend to be extremely healthy, these effects will not be influenced by confounding factors such as clinically relevant, age-related disease. Also, because the effects of spaceflight on immunity are temporary, generally returning to baseline levels post-flight, this will ultimately give us some insight into whether or not the effects of aging are, indeed, reversible.

Although many different aspects of immune function are influenced by both aging and spaceflight, these experiments will focus on B cells and the generation of antibodies. Specifically, we will use next generation RNA sequencing techniques to characterize the B cell Ig heavy chain in animals that are challenged with tetanus toxoid. The Ig heavy chain is the component of antibodies responsible for identifying specific sections of the tetanus toxoid, thereby coordinating subsequent immune responses. Comparisons will be made between the B cell repertoires generated on ISS and on Earth. The changes in B cell Ig heavy chain usage will provide indications whether the spaceflight environment affects the B cells that respond to antigen. This, in turn, will help us understand the impact of aging on B cell function.

We anticipate that we will obtain several data sets which have relevance to spaceflight and/or have general significance to understanding host immune responsiveness. These data include: 1) Data that show that spaceflight either maintains or alters the basic distribution of B cells based on Igh transcripts; 2) Information that demonstrates how immunization with an immunogen, commonly used in human medicine, alters the basic distribution of B cells based on Igh transcripts in the absence of the spaceflight variable; 3) Determinations how activation of B cells through Tlr9 receptors alters the basic distribution of B cells based on Igh transcripts.

We will ascertain if activation of B cells through Tlr9 receptors can reverse any impact that may occur during spaceflight.

Task Progress:

Project History & Review of Specific Aims
This project was submitted and funded in response to the NASA Research Announcement NNH12ZTT001N: Research Opportunities in Space Biology. This multidisciplinary project includes three independent investigators and two institutions (Loma Linda University, LLU and Kansas State University, KSU; in addition to NASA). The project proposed three different experiments aboard the International Space Station (ISS) to characterize different aspects of the antibody response. We proposed to immunize mice with the established tetanus toxoid (TT) vaccine. This would have enabled us to generate an antigen-specific antibody response in our mouse model while using a clinically-relevant antigen. Our goal was to determine the impact of the spaceflight environment on primary and secondary antibody responses. C57BL/6J mice would have been challenged with a TT vaccine either before launch or during their stay on the ISS, as appropriate. Blood and spleen were to be collected in orbit from mice 2-3 weeks after primary or secondary immunization. Vaccination and assessment schedules would have been matched in ground-based controls. The antibody response and the B cell immunoglobulin heavy-chain (IgH) repertoire would be determined in the splenic B cells from each treatment groups and compared among flight and ground treatment groups.

Due to the loss of astronaut work time assigned to Space Biology and =2 year delay in the transport of mice to the International Space Station, we were approved in 2015 to modify the aims of NNX13AN34G to include a Flight Definition Phase. Much of the last year has been completing the Flight definition phase ground based work and publishing the results.

1. Obtain spleen tissue/RNA from the mice flown on Rodent Research 1 (RR1) verification flight. We are currently analyzing spleen RNA from these tissues (10 mice; 5 flight and 5 ground controls). We have determined IgH usage of the spleen B cells in these mice (flight vs. ground controls). We also generated total RNASeq on the RNA to have a matching data set to the liver RNASeq that is being done by Gene Lab.

2. An antiorthostatic suspension (AOS) will be done to skeletally unload mice to simulate the skeletal unloading that occurs during spaceflight. In this protocol, mice were subjected to AOS. 2 weeks later the mice were immunized in 1 of 4 ways; PBS, CpG, tetanus toxoid (TT), and TT with CpG. Two weeks after immunization and continued AOS, we sacrificed the mice and collected a variety of tissues for further analysis. These are the same treatment groups that were proposed for our ISS studies. We used female C57BL/6J mice, 10-12 weeks old at the start of unloading. Independent variables were Tetanus Toxoid, CpG adjuvant, and hindlimb unloading. Blood and spleen were analyzed as part of NNX13AN34G. Assays at Loma Linda University (LLU) included blood, TT specific IgG, IgA, IgM and non-specific Ig levels. Kansas State University has completed analyzing all the IgH repertoire based on RNAseq and specific amplification of IgH usage.

Progress to Date

LLU Anti-orthostatic Suspension Data. The AOS was done to skeletally unload mice to simulate the skeletal unloading and stress that occurs during spaceflight (Chapes, Mastro et al. 1993) (Experiment 1 took place in March 2016 [5 mice per treatment group] Experiment 2 took place in May 2016 [5 mice per treatment group]). In this protocol, female, 8-week-old mice were subjected to AOS. Two weeks later the mice were immunized in 1 of 4 ways; PBS (buffer control), CpG only (adjuvant), tetanus toxoid (TT; antigen) only, and TT with CpG. Injections were done by intraperitoneal (i.p.) route. Two weeks after immunization and continued AOS (28 day total unloading), we sacrificed the mice and collected a variety of tissues. This protocol was followed so that when mice are flown to ISS in the future, we will have the AOS as a comparison group with parallel variables. Serum, spleen, liver and bone marrow were collected specifically to compare with our flight project.

Briefly, we were able to confirm that i.p. injections (as opposed to the more typical i.m. injection for vaccines) of TT resulted in an antibody response and that this response was augmented by an order of magnitude when mice were also given the CpG adjuvant. However, AOS had little impact on the ability to generate TT-specific antibodies, at least in the blood. AOS increased circulating corticosterone levels and a number of cytokines/chemokines, including GCSF, IL-1a, IL-5, IL-6, IP-10 and KC. RNAseq of the spleen, liver, and bone marrow from these mice is currently ongoing.

Note: In addition to these tissues, we collected heart, lung, brain, cerebellum, eye, skin, thymus, kidney and adrenals. All tissues were drop frozen in liquid nitrogen at the time of dissection. Our original intent (after considerable consultation with GeneLAB) was for these tissues to be used to generate a systemic, multi-organ reference data set that matched our spaceflight experiment. However, we have not yet identified funding for this project.

Assays on Collected Tissues

MiSeq analysis of the spleen was completed at KSU. Originally, we had intended to perform a similar analysis of the liver. However, as described in the recently approved modification to our related grant, NNX15AB45G (Collection of Immune/Stress-related Tissues from Mice Flown on ISS, PI: Chapes SK), this analysis has proven to be ineffective and we opted to analyze bone marrow instead.

In addition, we characterized changes in circulating corticosterone and found that AOS led to a significant increase. Furthermore, we noted a trend for the AOS response to be augmented in TT treated mice. We would like to confirm these measurements, as well as expand it to catecholamine levels, in adrenal tissues.

Although the tissue will be slightly different from those described in the original proposal, the aim of doing multiple tissue assessments/comparisons would still be accomplished. This would be a high-science return. We also believe that results from these studies will augment the data from the flight experiments, confirming the efficacy of AOS as a model for spaceflight for immune endpoints.

Analysis of Rodent Research 1 Liver Data. We analyzed GeneLab data sequencing data that was available from the GeneLab data base, GLDS-47, GLDS-48. The number of immunoglobulin reads (either heavy or light chain) were quite low, less than 1% of what we had obtained in preliminary analysis of spleen cells and tissue. We first attributed this low yield to short sequencing reads from the HiSeq sequencing method that the GeneLab did and provided in the database. Therefore, we sequenced both splenic (funded by another grant) and liver RNA from the Rodent Research 1 (RR1) validation flight, which was made available by the NASA Ames Research Center (ARC) (n=3 ground, n=3 flight). RNA was sequenced on the Illumina MiSeq platform (275bp, 2x) at the Kansas State Integrated Genomics Facility. Sequencing data was quality trimmed using CLC genomics workbench and mapped to reference libraries obtained from ImMunoGeneTics (IMGT) information system and the National Center for Biotechnology Information (NCBI) gene database. Datasets were submitted to the IMGT HighV-Quest tool for CDR3 characterization.

	The analysis of liver (quantified as part of the tissue sharing grant, NNX15AB45G) vs. spleen (quantified as part of THIS grant) reads indicated that liver immunoglobulin gene reads range from 0.26% of spleen immunoglobulin gene reads to 0.91%. The number of reads ranges from less than 100 to no greater than 622, depending on the sample. Although useful, publishable data from the spleen can be obtained (preliminary data presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research and presented in the GeneLab workshop by Chapes and Ward) the liver sequencing will not provide meaningful immunoglobulin gene data, even when the more comprehensive MiSeq next generation sequencing platform is used to sequence. The liver data won't be of any value and will not fulfill our goal of doing tissue comparisons. Other data from the MiSeq NGS data sets will be useful for other data mining attempts and will be made available to the GeneLab. Therefore, these sequencing runs were not a completely fruitless effort. Additional Opportunities in Collaboration with Dr. Elizabeth Blaber: ARC Anti-orthostatic Suspension Experiment. Due to the nature of the studies described above, we were unable to use the blood or spleen tissue collected in the LLU experiments described above to characterize the impact of unloading on WBC populations. Furthermore, the funding and flight requirements for our flight project did not allow us to compare male vs. female mice. Fortuitously, the opportunity arose to look at blood populations in a study with a design similar to our own. This project involved Elizabeth Blaber at Ames Research Center. The focus of Dr. Blaber's study was to study the impact of p21 on bone, muscle, and stem cells. Male and female mice were suspended for 15 or 30 days (the latter timepoint is very similar to the 28 day unloading in our AOS studies). At the appropriate timepoint, mice were sacrificed and tissues were collected immediately. We characterized blood populations using an ABC Vet automatic hematology analyzer. In addition, we collected adrenals, brain, cerebellum, eye, liver, and skin tissue and froze them in liquid nitrogen. Although we were primarily interested in the wild type controls for this study, we collected tissues from both wild type controls and the p21 knockout mice. Our analysis of the blood indicated that unloading decreased all of the major leukocyte population counts. This decrease was generally more pronounced in the lymphocyte subset than in the other populations. Female mice tended to have higher cell counts than males, but there was no significant unloading x sex interaction on any subset. Finally, counts were similar between wild type controls and p21 knockouts for all populations. Flight Status & Future Activities Our flight investigation is currently designated as Prime and is planned to fly on NG-11, scheduled to launch on 4/17/19. We have designated this experiment as Tetanus Antibody Response Developed In Space (TARDIS). At the request of NASA, this will combine experiments 1 and 2 of our original proposal, including: 1) a primary challenge in space, and 2) a primary challenge pre-flight + a secondary challenge in space. Our third flight experiment is currently under consideration for a future flight date. In support of this experiment, we are currently purchasing a computer to run the RNAseq analysis in house. Previously, this work was completed at KSU in Dr. Chapes' lab. However, with his retirement, all operations have been moved to LLU. Furthermore, we will be bringing over one of his PhD students to begin work in our lab as a post-doc starting in December 2018. Because our initial assessment of the tetanus toxoid was over a year ago, we are going to run a small verification test on a new lot of the toxoid to ensure that virulence will be as expected during the launch. This will be a small dose response study, focusing around the does optimized in previous tests. Endpoints will include TT-specific IgG levels. With the current flight date of 4/17/19, we expect to run this study sometime in December or January. Reference Chapes SK, Mastro AM, Sonnenfeld G, Berry WD. Antiorthostatic suspension as a model for the effects of spaceflight on the immune system. J Leukoc Biol. 1993 Sep;54(3):227-35. Review. PubMed PMID: 8371052
Bibliography Type:	Description: (Last Updated: 08/04/2025)
Abstracts for Journals and Proceedings	Chapes SK, Rettig T, Ward C, Nishiyama NC, Bye B, Hlavacek S, Pecaut MJ. "Assessment of mouse immunoglobulin gene-segment usage by mice in space flight and space flight models." Presented at the 2017 International Space Station (ISS) Research and Development Conference, Washington, DC, July 17-20, 2017. 2017 International Space Station (ISS) Research and Development Conference, Washington, DC, July 17-20, 2017. , Jul-2017
Abstracts for Journals and Proceedings	Rettig T, Ward C, Hlavacek S, Pecaut MJ, Chapes SK. "Analysis of the Antibody Repertoire of the Unimmunized, Unchallenged, C57BL/6Tac Mouse after Flight Aboard the ISS." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Bye BA, Ward C, Rettig TA, Hlavacek S, Pecaut MJ, Chapes SK. "Effects of Spaceflight on Immunoglobulin Variable Regions of Unchallenged Mice." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
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